1

Problem setup

$$\dot{x} = -0.1\,yz, \quad \dot{y} = 1.6\,xz, \quad \dot{z} = -0.75\,xy, \qquad u_0 = (0.5,\ 0.75,\ 1.0)$$

$t \in [0,\ 13.26446]$ · 27 windows · 1,536 points/window · float64

REPEAT FOR EACH WINDOW k = 1 … 27

2

PINN for window k

Input
t

Hidden
4×60 tanh

Output
$N_k(t) \in \mathbb{R}^3$

Hard IC
$\hat{u}_k = u_{0,k} + (t - t_k)\,N_k(t)$

3

ODE residual

$r_k = d\hat{u}_k/dt - f(\hat{u}_k)$, derivatives by autodiff

4

Causal residual loss

$$\mathcal{L}_i = \mathrm{mean}(r_i^2), \quad w_i = \exp\left(-\varepsilon \textstyle\sum_{j<i} \mathcal{L}_j\right), \quad \mathcal{L} = \mathrm{mean}(w_i \mathcal{L}_i)$$

$\varepsilon:\ 0.01 \to 0.1 \to 1 \to 10$

5

Adam optimization

learning rate × 0.9 every 1,000 steps

Stage complete?
$\min w > 0.99$
or 4,000 steps

NO

YES

All ε values used?

NO

Next ε
keep weights

next ε

YES

6

L-BFGS

≤ 1,000 iterations
unweighted residual MSE

7

Save window

checkpoint · history · prediction

Last window?
$k = 27$

NO

Next window
$u_{0,k+1} = \hat{u}_k(t_{k+1})$
copy weights, $k \leftarrow k+1$

next window

YES

8

Reference evaluation

DOP853 · 13,265 points · rtol 10^{-10} · atol 10^{-12}
used only for evaluation, never in training

9

Evaluation outputs

MAE · RMSE · max error · residual MSE · invariant drift · training cost